

UAV-Based Wildfire Detection and Localisation Using Deep Learning on the FLAME Dataset

Hong Anh Ta
bj24dsyu06

Nhi Nguyen Yen Ho
bs23dsy035

Duy Do
bj24dsyu07

Hung Viet Vu
bs23dsy027

I. PROBLEM DEFINITION

Early wildfire detection from unmanned aerial vehicle (UAV) imagery is an important computer vision problem because delayed or unreliable alarms can lead to greater fire spread, higher response cost, and more severe environmental damage. In this project, the problem is formulated as a two-stage UAV wildfire monitoring task using the FLAME dataset. The input is an aerial RGB image or video frame acquired from a UAV platform, and the system is expected to produce two outputs: (1) a frame-level prediction indicating whether fire is present, and (2) for frames predicted as fire-containing, a pixel-level segmentation mask identifying the fire region. This problem formulation follows the benchmark proposed by Shamsoshoara *et al.* [1], in which wildfire monitoring is studied through both fire/no-fire classification and fire segmentation.

The main challenge is not only to detect fire, but to do so reliably and efficiently in realistic outdoor conditions. Fire regions may be small relative to the full aerial frame, while visually complex backgrounds such as vegetation, smoke, water, snow, shadows, and bright ground surfaces can resemble fire and increase the risk of false alarms. This challenge is explicitly reflected in FLAME, where Shamsoshoara *et al.* [1] reported that the baseline classifier produced a higher false-positive rate than false-negative rate. Accordingly, the central research problem is to develop a UAV-based fire detection and localisation pipeline that can maintain strong recognition performance while reducing false positives and remaining computationally efficient for practical deployment.

II. MINI LITERATURE SURVEY

Recent wildfire vision research shows a progression from simple image-level recognition toward more practical systems that combine detection, localisation, and efficient deployment. The main starting point for this project is the FLAME dataset paper by Shamsoshoara *et al.* [1], which introduced a UAV wildfire benchmark with aerial RGB and thermal imagery and formalised two tasks: *fire/no-fire classification* and *fire segmentation*. Their benchmark used Xception for classification and a customised U-Net for segmentation. This paper is central not only because it provides the dataset, but also because it highlights a key limitation: the classification baseline showed a relatively high false-positive tendency. Reducing false alarms is therefore one of the main goals of the present research.

That benchmark motivates the first stage of this project, namely the establishment of representative baseline models for FLAME. Xception remains a meaningful reference because it is the original classifier used in the FLAME benchmark by Shamsoshoara *et al.* [1]. At the same time, later wildfire studies increasingly emphasise efficiency alongside accuracy. For example, Shi *et al.* [2] proposed an improved MobileNetV3-Small model for wildfire identification and reported strong classification performance with real-time processing capability. Their work is especially relevant because it shows that lightweight architectures are attractive for deployment in edge or onboard UAV systems with limited computational resources.

Beyond image-level classification, recent studies also stress the importance of segmentation and localisation. The FLAME paper already uses U-Net-style segmentation to localise fire regions after classification. More recent work suggests that segmentation should play an even more meaningful role in UAV-based wildfire monitoring. Pesonen *et al.* [3] investigated real-time wildfire segmentation for UAVs using smaller deployment-oriented models trained with larger teachers, showing that localisation can be made practical under onboard computational limits. This supports the idea that segmentation should be treated as an operational component of a full UAV fire alarm pipeline rather than a separate benchmark task.

Another recent development is the movement from plain binary masks toward boundary and fire-front interpretation. Li *et al.* [4] proposed a UAV-based pipeline consisting of fire segmentation, boundary tracking, and fire-front interpretation, arguing that wildfire analysis should provide more meaningful geometric information than only binary masks. Their study shows that segmentation can support richer spatial understanding of wildfire scenes, even if the present project is limited to localisation rather than full fire-spread modelling.

In addition, multimodal wildfire studies suggest that RGB imagery alone may not always be sufficient in difficult outdoor conditions. Yue *et al.* [5] proposed a multimodal deep learning framework combining RGB and thermal imagery for wildfire segmentation and showed that multimodal fusion can improve localisation accuracy and boundary quality. Although the present Stage 1 work does not yet incorporate full RGB-thermal fusion, this line remains important because FLAME also contains thermal data, making such an extension feasible in later stages.

This directly motivates the path of the present project: first establish representative Stage 1 baselines, and then improve the system through error analysis, lightweight model design, and alarm verification.

III. EXPLORATORY DATA ANALYSIS

A. Class Distribution

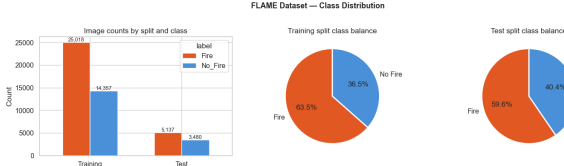


Fig. 1. Class distribution in the FLAME classification dataset.

This imbalance is not extreme, but it may still bias the classifier toward the Fire class. For this reason, augmentation is useful.

B. Image Dimension Consistency

A dimension audit showed that sampled images have a uniform size of 254×254 pixels with a square aspect ratio. This is beneficial because it simplifies preprocessing and model input preparation.

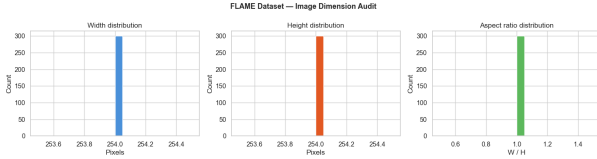


Fig. 2. Dimension audit of sampled FLAME images.

C. RGB Channel Analysis

The sampled pixel statistics show that Fire images have higher average RGB values and greater variation than No_Fire images. This suggests that Fire scenes are generally brighter and more visually diverse.

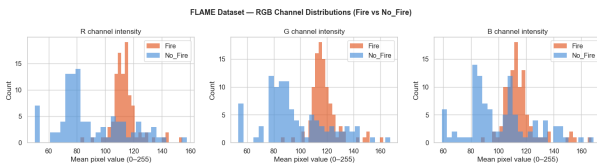


Fig. 3. RGB channel distributions for Fire and No_Fire classes.

However, there is clear overlap between the two classes. This means color alone is not sufficient for robust fire detection, and the classifier must learn richer visual patterns.

D. Visual Inspection

Visual inspection shows that fire often occupies only a small region of the UAV image, while the background may contain forest, smoke, water, snow, or low-light conditions. This makes the task challenging because the model must identify a small target under diverse scene conditions.



Fig. 4. Sample images from the FLAME dataset.

E. Statistical Summary

TABLE I
STATISTICAL SUMMARY OF THE FLAME CLASSIFICATION DATASET

Property	Value
Total images	47,992
Training images	39,375
Test images	8,617
Classes	Fire, No_Fire
Fire class share	62.8%
No_Fire class share	37.2%
Image resolution	254×254
Aspect ratio	1.0
Segmentation pairs indexed	0

F. Augmentation

The transformations include resizing, random cropping, horizontal and vertical flipping, and color jitter.

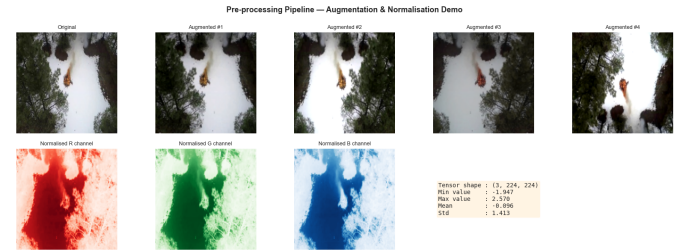


Fig. 5. Example augmentation outputs applied to a FLAME fire image.

IV. BASELINE AND PROOF OF CONCEPT

System Architecture: The proof of concept (PoC) is implemented as a unified PyTorch framework that integrates scene-level classification and pixel-level semantic segmentation within a modular single-notebook workflow. This design

ensures consistent data handling, preprocessing, and evaluation across tasks.

Model Architectures:

- **EfficientNet-B0 (Primary Baseline):** Selected for its compound scaling strategy. With approximately 5.3 million parameters, it provides a strong accuracy-efficiency tradeoff for real-time UAV inference.
- **U-Net (Segmentation):** Uses skip connections to preserve high-resolution spatial features, which are essential for delineating irregular wildfire boundaries.

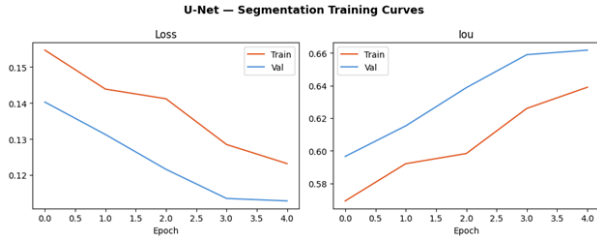


Fig. 6. Unet segmentation training curve

- **Benchmarking Variants:** MobileNetV3-Small and Xception are included as comparison models to benchmark performance against inference efficiency.

The Stage 1 baseline and benchmark models are summarised in Table II.

TABLE II
STAGE 1 MODEL SUMMARY

Model	Task	Role in Stage 1	Approx. Params	Status	Evaluation Metrics
EfficientNet-B0	Clf	Primary	~5.3M	Forward pass verified	Acc: 98.30%; AUC: 0.9982; F1: high
MobileNetV3-Small	Clf	Benchmark	~2.5M	Training cells available	Best val acc: 96.94%
Xception	Clf	Benchmark	~20M	Training vision cells available	Best val acc: 97.97%; val loss: 0.0672
U-Net	Seg	Core arch	~1.9M	Forward pass verified	Val IoU: 0.6617; val F1: 0.7957

FLAME Dataset — Sample Images per Class



Initial Findings: Preliminary experiments show that the classification models converge stably and achieve strong val-

idation performance, while the segmentation branch produces promising localisation results. These findings confirm that the Stage 1 pipeline is functional and suitable for later comparison, optimisation, and false-positive analysis.

REFERENCES

- [1] A. Shamsoshoara *et al.*, “FLAME: Fire luminosity airborne-based machine learning evaluation,” *arXiv preprint arXiv:2012.14036*, 2021.
- [2] G. Shi *et al.*, “Wildfire identification based on an improved MobileNetV3-Small model,” *Forests*, vol. 15, no. 11, 2024.
- [3] J. Pesonen *et al.*, “Detecting wildfires on UAVs with real-time segmentation trained by larger teacher models,” in *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision*, 2025.
- [4] J. Li *et al.*, “Intelligent segmentation of wildfire region and interpretation of fire front in visible light images from the viewpoint of an unmanned aerial vehicle,” *ISPRS Journal of Photogrammetry and Remote Sensing*, vol. 220, pp. 473–489, 2025.
- [5] T. Yue *et al.*, “A multimodal deep learning framework for accurate wildfire segmentation using RGB and thermal imagery,” *Applied Sciences*, vol. 15, no. 18, 2025.
- [6] S.-Y. Kim and A. Muminov, “Forest fire smoke detection based on deep learning approaches and unmanned aerial vehicle images,” *Sensors*, vol. 23, no. 12, p. 5702, 2023.
- [7] F. Taxpulat, “Aerial wildfire detection using image classification,” *CS231N Course Project Report*, Stanford University, 2024.
- [8] A. Agarwal, J. Gearon, R. Rank, and E. Chenevert, “Fighting fires from space: Leveraging vision transformers for enhanced wildfire detection and characterization,” *arXiv preprint arXiv:2504.13776*, 2022.
- [9] A. Shamsoshoara, “Fire-Detection-UAV-Aerial-Image-Classification-Segmentation-UnmannedAerialVehicle,” GitHub repository.